

Investor trade allocation patterns in stock markets

Kęstutis Baltakys^{a,*}, Juho Kanninen^a, Jari Saramäki^b, Mikko Kivelä^b

^a*Financial Computing and Data Analysis, Department of Computing Sciences, Tampere University, Finland*

^b*Department of Computer Science, School of Science, Aalto University, Espoo, Finland*

Abstract

In this paper, we statistically analyze how investors distribute their trading intensity to different securities in stock markets. We find that trade allocation distributions are distinctive between investors. More importantly, the patterns of trade allocation across securities are surprisingly persistent for each investor, even when there is turnover of securities in their portfolios. This suggests that even if investors have security-specific strategies, the securities can be replaced by others while the strategies remain. We also find that time constraints do not appear to limit investors' activeness: the larger the set of securities an investor trades, the more frequently she trades each security on average.

Keywords: Investor behavior, Trade allocation, Portfolios, Stock markets

1. Introduction

A human investor has a limited amount of time for gathering information for identifying trading opportunities and preparing and executing her trading decisions. Because attention is a scarce cognitive resource (Kahneman 1973), an investor must decide how to distribute their limited attention across different securities. In this paper, we introduce the concept of *Trade Allocation frequency-rank Distribution* (TAD), which quantifies how an investor's trading activity is distributed across different securities in a given time period. TADs allow us to study how investors distribute their trading

*Corresponding author. Tampere University/Group for Financial Computing and Data Analytics. Email kestutis.baltakys@tuni.fi.

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intensity across different securities in stock markets.² We study TADs empirically using an investor-level data set with more than 1 million individual investors. The data set contains every stock market transaction executed on the Helsinki Stock Exchange over 22 years. In this paper, we focus on individual retail investors and financial institutions, namely financial-insurance corporations, that traded in Helsinki Exchange, Finland.

Our approach can be visualized by building a trade allocation graph for each investor by linking the investor to the securities she has traded.³ An illustration of such a graph and the resulting TAD is shown in Figure 1A. Then, by using statistics on the TADs, we can analyze how investors distribute their trading intensity across securities over time. The identification of investors' TADs allows us to analyze their trade allocation mechanisms from several points of view. While it is known that retail investors have underdiversified portfolios (Barber and Odean 2013), as a minor result of our preliminary analysis, we find that investors distribute their trading intensity highly heterogeneously across their securities. This finding is in line with Mental Accounting (Das et al. 2010) which suggests that investors have sub-portfolios (Das et al. 2010), and also provides evidence that investors' rebalancing strategies can be sub-portfolio-specific. For example, an investor could apply a passive investment strategy to one sub-portfolio but an active strategy to another.

Second, and more importantly, we observe that TADs differ statistically between investors, while the way how each investor allocates trades across multiple securities is persistent in time. It takes surprisingly long, from eight to 15 years, for her to change the pattern of trade allocation considerably. The literature provides evidence about persistence in investor behavior but related to other aspects of trading, such as herding (Hwang and Salmon 2004), market inefficiencies and the behavior of informed investors (McMillan 2003), and the performance of mutual funds (Harless and Peterson 1998,

²Trade allocation should not be confused with asset allocation. When we discuss trade allocation, we analyze how investor distributes her trade days (events) across different securities, whereas asset allocation is about where to put money in the market. Importantly, an investor can trade a given security rarely with a passive strategy, while, compared to more frequently traded securities, this security can be one of the main components in her portfolio. In this case, the trade allocation of this security would be low, but asset allocation would be high.

³The investor trade allocation graph resembles so-called ego networks, where alter nodes are connected to an ego node. In many social network studies (see, for example, Arnaboldi et al. 2012, Saramäki et al. 2014), both egos and alters are persons, but in this paper, alters are securities, and hence the egos and alters are of different nature.

Busse and Irvine 2006).

Moreover, we analyze if the trade allocation is affected by the turnover of securities in the investors' portfolios. We find that the way how an investor distributes her trading intensity across the securities is statistically independent of the composition of her portfolio. This may indicate that investors do not ' earmark ' securities in their portfolio strategies. This finding is consistent with how people manage their social capital (Saramäki et al. 2014). Whereas a person gets new friends, some old friends either are replaced or receive less attention, preserving the overall attention distribution across the friends.

Third, we investigate the time constraints of human investors by asking if investors with large portfolios trade on average each of their securities less frequently. Intuitively, it is to be expected that retail investors are subject to time constraints. However, surprisingly, this is not what we find, even for investors having as large portfolios as *hundreds of securities*. We provide robust evidence that the larger the portfolio of an investor, the more frequently she trades each of her securities on average. This means that investors are able to increase their investment intensity with portfolio size. This result is shown to be consistent with data from different decades, and it is also robust to how the trading activity is measured. We also conduct a similar analysis with institutional investors and find no significant differences compared to the retail investors. Our explanation for this observation is that retail investors are able to scale their trading effort by changing their trading strategies from more informationally intensive to a strategy operated with technical rules.⁴

Finally, there is an interesting analogy with social systems where people have limited time and effort to initiate and maintain social relationships. Markets and societies are, at first glance, very different types of systems. Yet, both are driven by the choices of people who have to decide how to allocate their attention and resources. Recent studies on large-scale communication data have revealed characteristic patterns in people's strategies to invest in social relationships (Roberts 2010, Miritello et al. 2013, Saramäki et al. 2014). The trade allocation distributions resemble so-called social signatures, which are frequency-ranked distributions of social relationships whose strength is characterized by communication frequency (see Saramäki

⁴For example, a stock picker who trades relatively rarely could become a portfolio investor who rebalances the portfolio in terms of co-variances of the securities, thus trading each security more frequently in a larger portfolio with a cheap strategy.

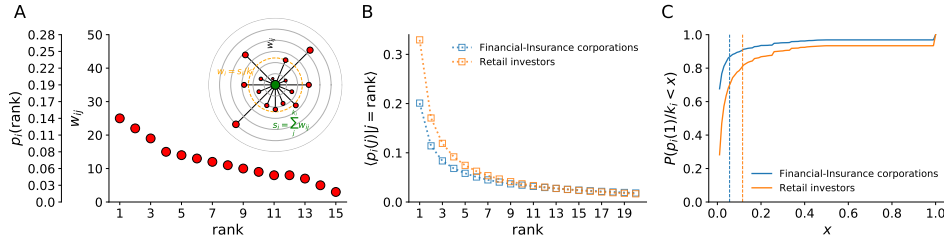


Figure 1: Heterogeneity in trade allocation and differences between retail investors and institutional investors. (A) A Trade Allocation frequency-rank Distribution (TAD) together with the underlying investor trade allocation graph (inset). In this graph, the central node stands for the investor, and the surrounding nodes represent securities traded by the investor. Link weights w_{ij} represent the intensity of trade. The TAD is constructed by ranking the investor's securities according to their link weights. For an investor, it represents a conditional probability distribution of ranked securities that the investor trades on a given day, given that she is active on that day. (B) Averaged annual TAD patterns for retail investors and financial institutions for the top 20 ranks. (C) Cumulative annual distributions of the slopes of the TADs, calculated as the ratio of trading intensity directed towards the first rank and the length of the pattern ($p_i^t(1)/k_i^t$) for retail investors and financial institutions. The respective mean values, shown as dashed vertical lines, indicate that retail investors emphasize their top securities the most. (D) The empirical cumulative distribution of the correlations between investor's trading intensity (in terms of the number of transactions) and the total number of transactions of the same securities in the market.

et al. 2014). Interestingly, we observe some striking similarities between how people manage their social capital and financial assets. Similarly to our trade allocation patterns, people have communication patterns that persist in time despite considerable turnover in their network. However, our result of increasing trade frequency with increasing portfolio size is in contrast to social networks, where it has been shown that people with larger networks dedicate on average less time to each of their social relationships Miritello et al. (2013) which has been associated with the effects of time constraints.

2. Materials and Methods

2.1. Data

Our data, provided by Euroclear Finland, consists of detailed daily information on all transactions⁵ in the Helsinki stock exchange from January

⁵Transactions represent executions of limit orders, not unexecuted limit orders.

1995 through December 2016, with 1.18 million active retail investors making 52.2 million transactions in 1877 securities (see column I in Tab. 1). We only use data on the top 5% ($N = 1884$) of the Helsinki stock exchange's most actively traded securities (in terms of trading days). This limits the effect of security liquidity (or the lack of it) on investor decisions. Key statistics about the traded securities are summarized in Tab. 2.

For the analyses of trade allocation patterns in Sections 3.1, 3.2, and 3.3, and the inter-trading times in Section 3.5, we restrict the data set to investors who have traded on ≥ 340 distinct trading days. The choice of 340 days is arbitrary and was chosen so that, in the restricted data set, we would get roughly 10 thousand retail investors (see columns III, IIIr, and IIIf in Tab. 1). To investigate the relationship between the number of traded securities and the total trading intensity in Sec. 3.4, we also include less active investors (see columns II and IIr in Tab. 1). The anonymized trade allocation pattern data set is available online⁶.

The number of trading days was used as a proxy for trading intensity in the main analysis because the number of transactions and total volume were not available after 2009. Particularly, after 2009, only net-volumes (total number of shares bought minus total number of shares sold) for each security were reported. Therefore the trading volumes or the number of transactions were not available for the whole period. Here we prioritized having 2010's involved in the analysis over the use of these alternative measures for investor trading intensity. Most importantly, the robustness checks show that the result do not significantly change when using these alternative measures for 1995–2009. Moreover, Figure 9 in Appendix shows that the bulk of the traded securities appeared from 2010 onwards.

The shareholder registration data we use in this paper is rather unique. There are few investor-level data sets covering such a long period and including all the trade-by-trade transactions. While the data set covers only Finnish securities, the results are generalizable to other markets too. This is because Finnish investors are quantitatively and qualitatively similar to investors in other developed markets (see Grinblatt and Keloharju 2000, Keloharju and Lehtinen 2021, Grant et al. 2022). The table shows that Finnish households are rather active in buying or sellign securities: out of whole population (around 5 million people), more than 1 million have traded at least once between 1995-2016. For example, just in 2015, there were 740,000 active distinct investors (see Keloharju and Lehtinen 2015).

⁶[10.6084/m9.figshare.12917882.v1](https://doi.org/10.6084/m9.figshare.12917882.v1)

	I	II	IIr	IIIf	III	IIIr	IIIIf
Panel A. Key data set statistics							
transactions	100.44 M	94.13 M	52.19 M	21.01 M	60.32 M	23.10 M	20.67 M
transaction days	43.33 M	40.93 M	33.20 M	2.27 M	16.94 M	11.62 M	2.16 M
investors	1.31 M	1.27 M	1.18 M	1,270	11,900	10,007	229
securities	37,613	1,884	1,877	1,595	1,884	1,803	1,582
Panel B. Number of traded securities over the whole period by an investor							
median	2	2	2	15	67	66	116
mean	6.91	6.35	6.22	39.87	72.36	70.17	133.95
max	9,484	991	352	991	991	352	991
Panel C. Number of traded securities in a year by an investor							
median	2	2	2	6	11	11	29
mean	3.69	3.60	3.47	16.60	14.32	13.41	37.46
max	2,962	470	133	470	470	132	470
Panel D. Annual investor's intensity s_i measured in trading days							
median	2	2	2	12	37	35	134
mean	9.29	9.10	7.96	231.67	85.93	68.95	658
max	51,965	36,827	4,089	36,827	36,827	4,089	36,827
Panel E. Annual investor's intensity s_i measured in transactions							
median	3	3	3	19	49	46	244
mean	21.54	20.93	12.51	2,143	305.98	137.08	6,311
max	1.02 M	1.02 M	104,253	0.88 M	1.02 M	104,253	0.88 M
Panel F. Annual investor's intensity s_i measured in Euro-Volume							
median	4,696	5,029	4,667	332,924	198,241	164,709	12.52 M
mean	0.87 M	0.90 M	82,368	153.54 M	17.93 M	1.17 M	451.32 M
max	126.15 B	126.15 B	2.90 B	40.10 B	126.15 B	2.90 B	40.10 B
Panel G. Sections where datasets are used							
Sections			3.4	3.4		3.1, 3.2, 3.3, 3.5	3.1, 3.2, 3.3

Table 1: Transaction data set statistics. Columns: I—complete data set; II—5% of most traded securities; IIr—retail investor, and IIIf—financial-insurance corporation subset of II; III—investors active on at least 340 days; IIIr—retail investor, and IIIIf—financial-insurance corporation subset of III. The numbers in columns with retail investor data (IIr and IIIr) and financial-insurance corporation data (IIIf and IIIIf) do not sum up to columns II and III as they include investors from other sectors too. Panel A describes the key statistics of the data sets: the overall number of transactions, transaction days, investors, and securities. Panel B describes the distribution of the number of traded securities by an investor over the whole data set. Panel C describes the distribution of the number of traded securities in a year by an investor. Panels D, E, and F describe the distribution for the annual investor's trading intensity as measured by trading days, transactions, and euro-volume turnover.

	# investors per year	# transactions per year	Euro-volume per year	Returns	Volatility
mean	2,402	54,398	3.76 B	-0.24	0.46
std	9,984	292,521	35.21 B	3.17	0.73
25%	49	534	1.57 M	-0.81	0.20
50%	174	2,370	14.10 M	0.00	0.30
75%	1,160	12,003	218.52 M	0.76	0.46

Table 2: Annual statistics about the securities, including all marketplace transactions in Helsinki Stock Exchange data set II (see Table 1). The column about number of investors come from the data over the whole period from 1995 to 2016 behind the 1884 securities. The rest of the columns discuss the complete trading data from 1995 to 2009 behind 635 securities for which we have price data. Returns and Volatility observations are calculated over 30-day periods and annualized.

2.2. Frequency-count distributions for investor trade allocation

For each investor, we compute *investor trade allocation frequency-count distributions*, TADs, by ranking the securities that investor i has traded in year t according to trade frequencies across the securities and then calculate the fraction of trading intensity given to each security as a function of its rank. The resulting TAD describes how each investor distributes her total trading intensity from the most to the least frequently traded security. At the same time, it represents a conditional probability distribution of securities investor trade on a given day, given that she is active on that day.

In a trade allocation graph (see Fig. 1A), the focal investor i is connected to security j if she has traded j within the observation period. The degree k_i is the total number of securities she has traded during this period. The links in the trade allocation graph are weighted by w_{ij} , defined as the number of days where investor i has carried out at least one transaction with security j . The sum of these weights, $s_i = \sum_j w_{ij}$, is a proxy for investor i 's total trading intensity (see Appendix for alternative measures of trading intensity).

We build investor trade allocation graphs from the trading data, which are investor-centric subgraphs of a bipartite graph that connect investors to their traded securities (see Fig. 1A). Over the whole 22 years, the median investor trades only two securities, while the average investor trades 6.22 securities, and the most active trades 352 securities. Alternatively to the number of trading days, the number of transactions or the total turnover could have been used to define the edge weights. Since these measures are highly correlated, we only report results for the trading-day measure.

Since the majority of investors are passive traders and the goal is to

investigate how the trading intensity is distributed across securities, we limit the analysis to the top 10 007 most active investors in the period from 1995 to 2016. In this reduced data set, the median individual retail investor trades 66 securities, the average 70.17, and the most active investor 352 securities as before. To construct the annual trade allocation patterns for investor i , we compute for each security the fraction of trading $\{p_{ij} = w_{ij}/s_i\}, j = 1 \dots k_i$ and rank the securities in ascending order:

$$\max_j p_{ij} = p_i(1) \geq \dots \geq p_i(n) \geq \dots \geq p_i(k_i) = \min_j p_{ij}, \quad (1)$$

where $p_i(n)$ denotes the n th largest observation of p_{ij} . The TAD for investor i is denoted as $P_i = \{p_i(1), \dots, p_i(k_i)\}$. That is, investor i 's TAD P_i is a non-increasing sequence of the share of total trading allocated to securities ranked from the most to the least actively traded.⁷

In Section 3.2, we analyze the similarity of TADs between consecutive periods and also between different investors, as well as the persistence of the TADs. We use *Jensen–Shannon divergence* (JSD) as a proxy for the distance between two TADs. JSD measures how similar the pairs of TADs are, defined as

$$\text{JSD}(P_1, P_2) = H\left(\frac{1}{2}P_1 + \frac{1}{2}P_2\right) - \frac{1}{2}[H(P_1) + H(P_2)], \quad (2)$$

where P_1 and P_2 are two investor TADs and $H(P_i)$ is the TAD entropy defined as

$$H(P_i) = -\sum_{r=1}^{k_i} p_i(r) \log p_i(r). \quad (3)$$

3. Results

3.1. A small number of securities receive a large share of an investor's trading intensity

As a preliminary and descriptive analysis, we first investigate how frequently investors trade their different securities. Heterogeneity in this frequency may result from stock-specific trading strategies. As an example, an investor may consider some of her securities as speculative assets, trading

⁷For more information about frequency-count distributions, see, for example, (Brookes and Griffiths 1978).

them more frequently with a short-term investment, or some securities may be prioritized above others. Seminal research in (Thaler 1999) and (Benartzi and Thaler 2001) provides evidence that mental accounting influences individuals when they organize, evaluate, and keep track of financial activities. With mental accounting in mind, we frame our question as *do investors place their securities into distinct categories that receive different amounts of trading?*

To see if there is a common pattern in how investors distribute their trading over securities, let us take a look at Figure 1B, which reveals that, on average, a small number of securities receive a large share of an investor's trading intensity, while the rest are much less traded. This pattern resembles the way how people distribute their attention to other people when maintaining social relationships (Saramäki et al. 2014). The strong emphasis on a few securities is much more distinct for retail investors than for financial and insurance corporations, whose average TAD shapes are shown for comparison in Fig. 1B. The average fraction the retail investors trade the most frequently traded security is $\langle P_i^t(1) \rangle = 0.330 \pm 0.246$ and to the top 5 securities $\langle \sum_{n=1}^5 P_i^t(n) \rangle = 0.728 \pm 0.226$.⁸ Correspondingly, $\langle P_i^t(1) \rangle = 0.201 \pm 0.210$ for financial and insurance corporations while $\langle \sum_{n=1}^5 P_i^t(n) \rangle = 0.504 \pm 0.265$ for financial and insurance corporations.

We also calculate the cumulative distributions of the slopes of TADs, defined as the ratio between the trading intensity directed to the first rank in the TAD and the length of the TAD ($p_i(1)/k_i$). This provides a way of comparing retail investors and institutional traders while accounting for their different TAD lengths. We find that the mean slope for retail investors is higher than for institutional investors, further supporting the hypothesis that retail investors choose different decision-making strategies across the set of their traded securities (see Fig. 1C). Moreover, there is an apparent downward drift in the average fraction of trading intensity given to the most traded securities from 1995 to 2000, followed by a stable trend from 2005 onward (see also, Fig. 6 in the Appendix). During the first years of the observation period, retail investors traded their top securities relatively more frequently than in the modern era of financial markets, while institutional investors have maintained a similar level throughout the period. These results confirm the existing evidence that investors differ in their stock market participation and risk taking (Grinblatt et al. 2011, Häusler et al. 2018).

⁸Here $\langle \cdot \rangle$ is the Reynolds operator used for averaging.

Moreover, we analyze how the characteristics of traded securities are related to investors' TADs. First, we check if the TADs reflect market-wide security trading intensities by analyzing the correlations between investor's trading intensity (in terms of the number of transactions) and the total number of transactions of the same securities in the market. If the correlation is one for a given investor, then she weights different securities in the same way the markets do (in terms of the number of transactions). Here the length of the TAD must be at least four, i.e. an investor must have been trading at least four securities. We observe that around 40% of investors have a negative correlation, and the average correlation is 0.22 ± 0.42 . Moreover, the median is 0.17, and the 75th quantile is 0.55. This shows that investors' TADs cannot be reliably approximated from the aggregated market-wide transaction data.

Additionally, we provide information about the correlations between TADs' entropies and measures that describe how securities are traded at the market level. Specifically, we use means for security volatility, return, euro volume, and the total number of transactions. From Table 3 we see that the correlation between the TADs' entropy and the measured characteristics of the securities is rather low. The highest (in absolute value) is the correlation against the number of transactions, -0.23. That is, the more an investor, on average, is trading her securities, the more concentrated her trading strategy is. This could be related to the day-trading, which requires more effort, compared to less-active naive portfolio rebalancing strategies, such as $1/N$ (for $1/N$ strategy, see, e.g., DeMiguel et al. 2009).

Table 3: Correlation matrix for investors' TAD entropy and mean values of security volatility, return, euro-volume, and total number of transactions. The bolded numbers show the correlations against the TAD entropy.

	TAD entropy	Return	Volatility	Eurovolume	# of transactions
TAD entropy	1.00	0.09	-0.10	0.02	-0.23
Return	0.09	1.00	-0.44	0.00	-0.06
Volatility	-0.10	-0.44	1.00	-0.01	-0.17
Eurovolume	0.02	0.00	-0.01	1.00	0.01
# of transactions	-0.23	-0.06	-0.17	0.01	1.00

3.2. Trading allocation distributions are persistent and heterogeneous across investors

We now move from the analysis of aggregated TADs to the level of individual investors. The trading patterns of investors are dynamic, and this is reflected in their trade allocation graphs—there are changes in the

investors' portfolios as well as in the intensities with which their securities are traded. We ask whether these micro-level changes affect the overall pattern that the TADs capture, and whether there are persistent individual differences in the shapes of TADs, reflecting individual trading strategies.

We want to investigate if investor's TAD is more similar to her future TAD than to the TADs of other active investors in the current time period. The distance between investor i 's TADs in two consecutive windows t and $t + 1$ is denoted as $d_{\text{self}}^{t,t+1}(i) = d_{ii}^{t,t+1} = \text{JSD}(P_i^t, P_i^{t+1})$, whereas the reference distances between investor i 's own TAD and the TADs of other investors during the same period t are denoted as $d_{\text{ref}}^{t,t}(i, j) = d_{ij}^{t,t} = \text{JSD}(P_i^t, P_j^t)$, where $j \neq i$. We define the set of all TAD self-distances in consecutive periods as $d_{\text{self}} = \{d_{\text{self}}^{t,t+1}(i); i \in I^{t,t+1}, t = 1 \dots 21\}$ and the set of all reference distances as $d_{\text{ref}} = \{d_{\text{ref}}^{t,t}(i, j); i, j \in I^{t,t}, i \neq j, t = 1 \dots 21\}$, where I^{t_1,t_2} is the set of all investors who were active during periods t_1 and t_2 . To this end, we compare the distributions of all self-distances d_{self} , to the reference distributions of all between-investor distances d_{ref} .

Averaged over all investors, the mean self-distance for consecutive periods is $\langle d_{\text{self}} \rangle = 0.087 \pm 0.088$ for retail investors and $\langle d_{\text{self}} \rangle = 0.058 \pm 0.082$ for financial institutions. In contrast, the average distance to other investors with $\langle d_{\text{ref}} \rangle = 0.158 \pm 0.123$ for retail investors and $\langle d_{\text{ref}} \rangle = 0.170 \pm 0.146$ for financial institutions. That is, investors' self-distances are statistically lower than distances between different investors, were they retail investors or financial institutions. This can be seen also from Figure 2, which shows the distributions of self- and reference distances for retail investors (Plot A) and institutions (Plot D). These findings have two implications: first, an investor's way to distribute trading intensity among securities in time is relatively consistent and persistent. Second, investors have relatively different trade allocation distributions. The difference between the average reference distances is larger than the one observed for social signatures (Saramäki et al. 2014). That is, people are more heterogeneous in terms of their trade allocation to stocks than their social attention allocation to their friends.

Moreover, there is long-term persistence in investors' trade allocation mechanisms as well, which we verify by two tests. In the first test, we calculate the self-distances of TADs for different annual lags L , $d_{ii}^{t,t+L}$, which we compare against contemporaneous reference distances $d_{ij}^{t,t}$ (see, Fig. 2B for retail investors and 2E for financial institutions). By this way, we quantify how long it takes, on average, for an investor's TAD not to be statistically different to other investors' TADs from the beginning of the comparison. The alternative hypothesis for lag L is that investor's current TAD to her

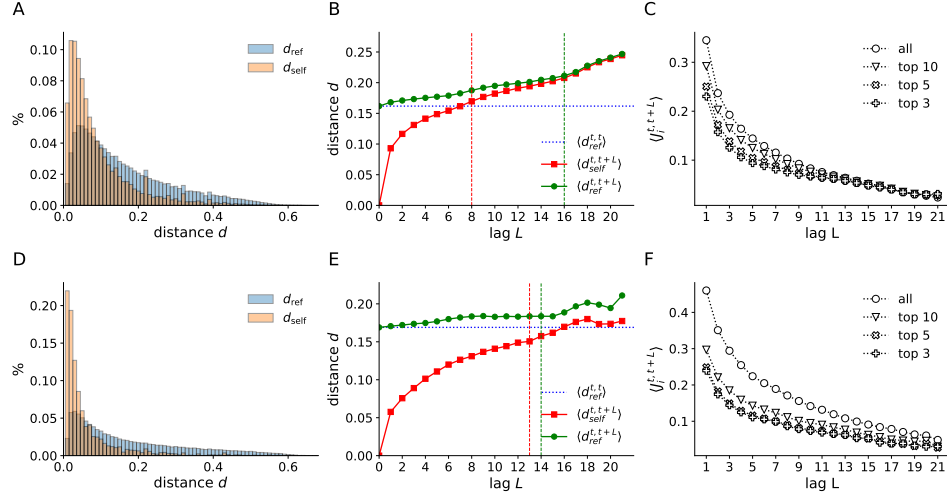


Figure 2: Persistence of TADs and stock turnover. (A) The distributions of self-distances d_{self} , defined as the distance between investor's TADs in two consecutive years, and reference distances d_{ref} , defined as the distance between investor's TAD and other investors' TADs in the same year. The distributions are pooled over the 21 pairs of consecutive years. The one-side Kolmogorov-Smirnov p-value is 0, which provides evidence that $\text{CDF}(d_{\text{self}}) > \text{CDF}(d_{\text{ref}})$, indicating that the values of d_{self} are lower than the values of d_{ref} . (B) The mean self-distance $\langle d_{\text{self}}^{t,t+L} \rangle$ for different time lags L in years (red), mean reference distance $\langle d_{\text{ref}}^{t,t} \rangle$ with zero lag (dashed blue line) and mean reference distance $\langle d_{\text{ref}}^{t,t+L} \rangle$ for different lags L (green). The vertical dashed lines indicate the lags at which for the first time the respective null hypotheses have not been rejected by one-side two-sample t-test at significance $\alpha = 0.0001$. (C) The averaged decay in Jaccard similarity between top 3, 5, 10, and all securities in yearly TADs with different lags $\langle J_i^{t,t+L} \rangle$. (D–F) are corresponding plots for financial institutions (Kolmogorov-Smirnov p-value is 0 also in plot D).

own TAD in the future is more similar (lower distance) than if we compare it to other investors' current TADs. Retail (institutional) investor's current TAD is significantly closer to her future TADs over the next eight (13) year period compared to other investors' current TADs. (Welch's one-tail t-test at $\alpha = 0.0001$, see red vertical line in Fig. 2B and 2E). Secondly, we compare self-distances and reference-distances for the same annual lags L , i.e. $d_{ii}^{t,t+L}$ and $d_{ij}^{t,t+L}$, where $j \neq i$ and $L = 1 \dots 21$. Here the alternative hypothesis for lag L is that investor's current TAD to her own TAD in the future is more similar (lower distance) than if we compare it to other investors' TADs in the future. Here retail (institutional) investor's current TAD is significantly closer to her future TADs over the next 16 (14) year period compared to the distance of other investors' TADs now and in the future (Welch's t-test

at $\alpha = 0.0001$, see green vertical line in Fig. 2B and 2E).

When we compare the results between retail investors and financial institutions (Fig. 2B vs 2E), we observe that TADs are not less diverse between financial institutions than between retail investors. This is quite surprising, because one could assume that financial institutions, and especially insurance companies, would have more homogeneous trading strategies and thus more similar TADs because of regulation and portfolio constraints. Also, an institution presents a pool of its investors, but, at the same time, the TADs of institutions are at least as distinctive as those of individual retail investors. This indicates that institutions have distinctive strategies, implemented by multiple traders, on how much different securities are traded.

3.3. Trade allocation distributions are not affected by changes in portfolio composition

If the TADs show persistence, how dynamic are the underlying portfolio compositions? To answer this question, we quantify changes in the set of traded securities between two different periods with the Jaccard similarity index. For investor i who has traded securities in periods t_1 and t_2 , we denote the Jaccard index as $J_i^{t_1, t_2} = J(S_i^{t_1}, S_i^{t_2})$,

$$J(S_i^{t_1}, S_i^{t_2}) = \frac{|S_i^{t_1} \cap S_i^{t_2}|}{|S_i^{t_1} \cup S_i^{t_2}|}, \quad (4)$$

where S_i^t is i 's set of traded securities in period t . The behavior of the Jaccard index for increasing time lags is shown for different numbers of top-ranked securities (3, 5, 10 and the whole portfolio) in Figure 2C for retail investors and 2F for financial institutions. There is a monotonous decrease in the portfolio similarity, which indicates that, on average, the investors' portfolios become increasingly different over time. This is, of course, in part due to the changing investment landscape (see Fig. 9 in the Appendix).

To test if changes in TADs are related to changes in investors' portfolios, we investigate the association between TAD differences, quantified with the self-distance $d_{\text{self}}^{t, t+1}(i)$, and changes in portfolio composition, quantified with the Jaccard index $J_i^{t, t+1}$. We are particularly interested in the effects of changing the components of the portfolio rather than changes in the size of the portfolio. For that reason, we remove the effects of differing portfolio sizes and to focus on the replacements of the portfolio components. We do this by computing these measures for the joint length ($k_i^{t, t+1} = \min(k_i^t, k_i^{t+1})$) of the TADs and associated sets of securities. We find that the correlation between the portfolio similarities and TAD changes

practically disappears, as $\rho(\tilde{d}_{\text{self}}^{t,t+1}, \tilde{J}_i^{t,t+1}) = 0.00967$ for retail investors and $\rho(\tilde{d}_{\text{self}}^{t,t+1}, \tilde{J}_i^{t,t+1}) = -0.17$ for financial institutions, where $\tilde{d}_{\text{self}}^{t,t+1}(i)$ and $\tilde{J}_i^{t,t+1}$ are the modified self-distance and Jaccard index, respectively.⁹ This indicates that the shape of the TAD is statistically independent of changes in portfolio composition. That is, the way how an investor distributes trading intensity over her securities remains the same regardless of changes in the portfolio.¹⁰ This is consistent with findings for personal networks (Saramäki et al. 2014).

One can ask if our findings are driven by investors who frequently change the set of scarcely traded securities but constantly keep the most traded securities in the portfolio. By construction, securities that an investor seldom trades, say once every two years, rarely show up in the investor's TADs. That is, there can be relatively high turnover for less traded securities between different estimation windows. Therefore, it is possible that there are changes in the portfolio composition while the TADs remain the same, potentially yielding low correlation between $\tilde{d}_{\text{self}}^{t,t+1}(i)$ and $\tilde{J}_i^{t,t+1}$. For a robustness check, we further investigated this problem and replicated our analysis with top-5 ranked securities only. That is, for a given investor, we excluded all the other securities, but not the top-5 ranked securities in her TAD (note that top-5 securities are investor and period specific). In that way, we ignored those securities that an investor did not trade often. By these settings, we find that the correlation between the portfolio similarity measured via the Jaccard index and the change in portfolio shape measured via JSD is -0.11 for retail investors and -0.187 for Financial-Insurance corporations. Therefore, the results are pretty similar even if the analysis is run with the most traded securities only. This confirms that the low correlation is not a

⁹To calculate the correlation between portfolio composition and self-distances, we modify the TADs and calculate self-distances $\tilde{d}_{\text{self}}^{t,t+1}(i)$ and Jaccard indexes $\tilde{J}_i^{t,t+1}$ due to the difference in TAD lengths as follows. For each investor i , we first determine the shared length of both TADs $k_i^{t,t+1} = \min(k_i^t, k_i^{t+1})$.

Given that $k_i^t > k_i^{t+1}$, TAD P_i^t is modified as $\tilde{P}_i^t = \{\tilde{p}_i^t(1), \dots, \tilde{p}_i^t(k_i^{t,t+1})\}$, where $\tilde{p}_i^t(n) = p_i^t(n) / \sum_{j=1}^{k_i^{t,t+1}} p_i^t(j)$ and the distances are calculated between TADs $\tilde{P}_i^t$ and P_i^{t+1} , $\tilde{d}_{\text{self}}^{t,t+1}(i) = \text{JSD}(\tilde{P}_i^t, P_i^{t+1})$. One can similarly get $\tilde{d}_{\text{self}}^{t,t+1}(i) = \text{JSD}(P_i^t, \tilde{P}_i^{t+1})$ when $k_i^t < k_i^{t+1}$. The idea is that we trim the longer TAD to match the length of the shorter TAD and re-normalize the longer TAD. Similarly, for the modified Jaccard index we modify the portfolios observed in the both periods by taking into consideration only the top $k_i^{t,t+1}$ securities.

¹⁰For the robustness check, we did not modify the lengths of the TADs and we found that for the non-truncated TADs, the correlations are negative.

result of the turnover of the rarely traded securities in investors' TADs.

3.4. *Trading intensity increases with the number of traded securities*

Next, we ask if investors with larger portfolios trade each of their securities less because of possible time constraints. Particularly, we investigate the relationship between the number of traded securities k_i and the average trading intensity per security, $w_i = s_i/k_i$. The results are shown in Figure 3A for retail investors and 3C for financial institutions, which show results over the whole 22 year period as well as two five-year periods. Surprisingly, we observe a monotonically increasing average intensity $\langle w_i | k_i = k \rangle$ with respect to the number of traded securities. That is, investors with larger portfolios trade each of their securities more and hence show higher average trading intensity, as shown in Figure 3A and C. While the advent of algorithmic trading would seem like a plausible explanation for the observation, surprisingly, the results are highly consistent between data samples from different periods. Even in 1995–1999, when trading technologies were less developed compared to 2012–2016, the average trading intensity grew monotonously in the number of securities.

While a similar trend of increasing communication activity with increasing contact network size is also found in social networks (Miritello et al. 2013), social networks also display saturation: once the number of contacts is large enough, communication effort per connection saturates. This has been associated with both time and cognitive constraints. However, for trading, we do not observe saturation in trading intensity at the population level (see Figure 3A and 3C). The reason for the lack of saturation might be that individuals who trade in more securities are more professional and simply have more time for trading (perhaps not having another job at all). In fact, investors should have time for trading exponentially with respect to the portfolio size: they do not only trade higher number of securities but each security more frequently. Another alternative is that decision-making strategies with large portfolios require less effort, yielding higher trading intensity per security. Both alternatives are associated with higher financial savviness. In principle, it is also possible that the number of actively traded securities in the Helsinki stock exchange is too low for observing a saturation point. However, this explanation is not very plausible, as the number of securities traded by a single individual is as large as 352 in our data. This is a considerably large set of securities, even for a pure portfolio investor.

Further, we analyze this question at the investor level. For each investor i we calculate the relative changes in the number of traded securities, $\log(k_i^{t+L}/k_i^t)$, and in the average trading intensity per traded security

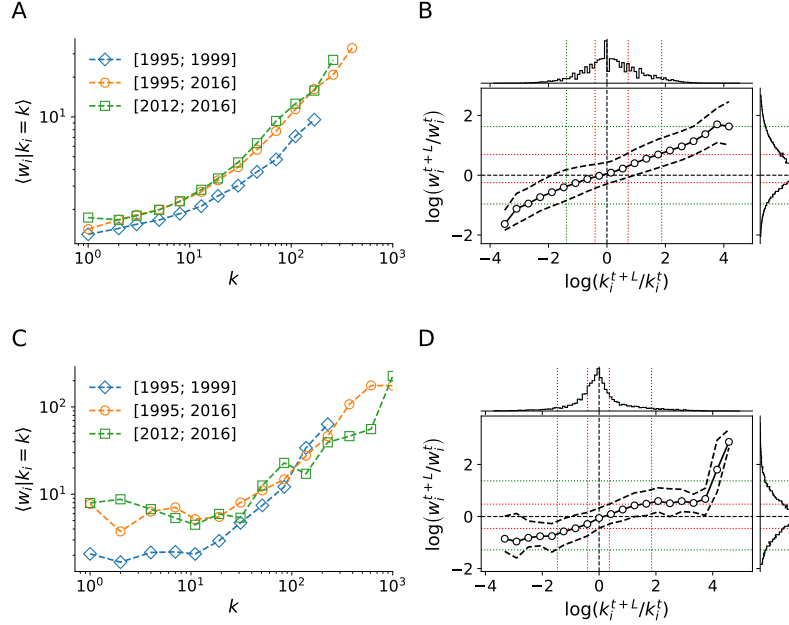


Figure 3: Variation of trading intensity. (A, C) The average trading intensity w_i with respect to the number of securities traded k_i for retail investors and financial institutions, respectively. The three curves represent three different time intervals. (B, D) Relationship between the relative annual change in the number of traded securities k_i^{t+L}/k_i^t and the median of the average trading intensity per traded security w_i^{t+L}/w_i^t for retail investors and financial institutions, respectively. Black dashed lines around median stand for the first and third quartiles. Red vertical and horizontal dotted lines stand for the first and third quartiles over all data points, while the green lines stand for the 5th and 95th percentiles. Histograms illustrate marginal distributions of the variables in respective axes.

$\log(w_i^{t+L}/w_i^t)$ between years t and $t+L$. We find that at the population level, the median of $\log(w_i^{t+L}/w_i^t)$ increases with respect to $\log(k_i^{t+L}/k_i^t)$ (see Figure 3B for retail investors and 3D for financial intuitions). This observation indicates that the more an investor increases (decreases) the number of securities in her portfolio, the more the level of trading per security increases (decreases). At the level of individuals, this positive relationship holds for more than 50% of investors, while only less than 30% of investors have a saturation point (see Table 5 in Appendix). A possible interpretation is that individual retail investors as well as financial institutions adjust their trading strategies over time: they may vary the overall level of effort directed towards trading as well as shift between decision-making strategies

that require different amounts of effort.

3.5. Inter-trading times are heavy-tailed

Finally, we investigate if the way investors time their trades is in line with how social interactions and other human activities are typically timed. More precisely, we study the distribution of inter-trade times. Heavy-tailed inter-event time distributions and bursty time series have been observed for a multitude of human behaviors, including communication in social networks. These inter-event time distributions in human activities have been explained by a queuing process model where recently executed tasks receive more attention and are more likely to be repeated (Barabasi 2005, Vázquez et al. 2006).

Burstiness denoted as B is defined as (Goh and Barabási 2008):

$$B = \frac{(\sigma_\tau/\mu_\tau - 1)}{(\sigma_\tau/\mu_\tau + 1)} = \frac{(\sigma_\tau - \mu_\tau)}{(\sigma_\tau + \mu_\tau)}, \quad (5)$$

where μ_τ is the average and σ_τ is the standard deviation of the inter-event time distribution of a sequence of trades. Memory coefficient denoted as M is defined as (Goh and Barabási 2008):

$$M = \frac{1}{n_\tau} \sum_{i=1}^{n_\tau-1} \frac{(\tau_i - \mu_1)(\tau_{i+1} - \mu_2)}{\sigma_1 \sigma_2}, \quad (6)$$

where τ_i and τ_{i+1} are the i th and $(i+1)$ th inter-event times, and μ_1 and μ_2 are the means of all τ_i and τ_{i+1} values. Similarly, σ_1 and σ_2 are the standard deviations of those values.

Figure 4 shows the distribution of normalized node activation inter-event times for four different investor activity groups. We can observe a data collapse, similar to ones observed in social interaction networks (Candia et al. 2008, Kivela and Porter 2015), for different investor groups as well as bursty behavior. The scatter plot in the inset of the same figure shows the relationship between the burstiness coefficient and the memory coefficient (Goh and Barabási 2008) for the inter-event times. The burstiness lies in the typical range of human activities, but the memory coefficients are on average higher (Goh and Barabási 2008).

4. Robustness checks and additional analysis

In this paper, we measure an investor's trading intensity by the number of distinct days when the investor i has traded security j , w_{ij} . Alternatively,

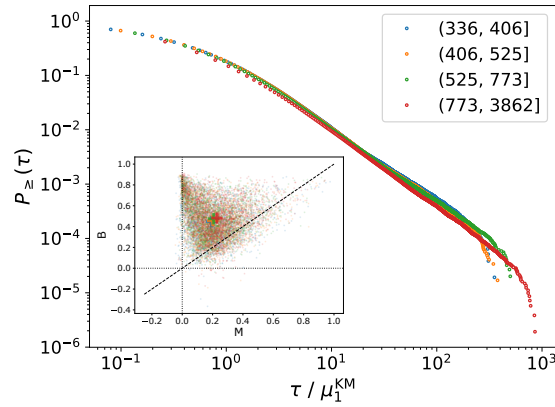


Figure 4: Node activation inter-event time distribution for four investor activity groups based on the number of distinct days they have traded, e.g., blue dots correspond to the investors who were active on between 337 to 406 distinct days in the analysis period. The inset shows the scatter plots for the same investor activity groups between the burstiness and memory coefficients for the inter-event times.

trading intensity could be measured by the number of transactions investor i has executed with security j , or the total volume i traded on security j (i.e. the sum of the volumes of all buy and sell transactions on security j). However, all three measures are highly correlated and we present results only for one of them. In particular, the average correlation between day and transaction counts is 0.90, while the average correlation between day count and the turnover measure is 0.77, and finally, the average correlation between the transaction count and the turnover measure is 0.78. For a robustness check, we run the results with alternative measures and as shown in Appendix, the results are very consistent across all the measures (see Fig. 7 in Appendix).

Moreover, in Appendix, we additionally analyze how an increase in an investor's total trading intensity is associated with the trade allocation. That is, if an investor has capacity to trade more often, does she distribute the extra capacity evenly across the securities, or allocate it only to the most important ones? We find that an increase in the total trading intensity of an investor is associated with a more diversified trading intensity to securities with flatter TADs (see Fig. 8 in Appendix). A possible interpretation is that frequent traders rationally aim to diversify their portfolios.

Third, investor's transactions can be driven by liquidity or other external, non-security-specific reasons, especially when it comes to sell-transactions.

For that reason, in Appendix, we provide results separately for buy and sell transactions. However, we find that the main results of this paper are consistent between buy and sell transactions.

Finally, one could suspect that the JSD measurement might be dominated by the difference in length of the two TADs, and for this reason, the largest portfolio in each pair could be truncated before computing the JSD. For the robustness check, along with Figures 2A and 2D, we calculated the Kolmogorov-Smirnov (KS) statistic between the self and reference distances with the truncation of the TADs so that the self and reference TADs include only the common number of traded securities. For the retail investors, the p-value of a one-sided KS test with truncation is 3.43e-317 (without truncation 0) and for institutions, p-value is 9.79e147 (with truncation 0)¹¹ This confirms, that the results are not dominated by the lengths of TADs. Nevertheless, the truncation of the TADs is not necessarily a good idea. The TAD entropy measures the investor's trading style concentration with respect to the securities. The underlying assumption can be that it is the investor's own decision not to trade too many securities. From this point of view, it is important to have a measure that does not only analyze the steepness of the TAD but also its length.¹²

¹¹With the data on retail investors, the distributions are not for all possible combinations of signatures – instead 1000 signatures are randomly sampled for every pair of consecutive years. That makes 21,000 (21×1000) observations for self-distance distributions and 10,489,500 ($21 \times (1000 \times 999/2)$) observations for the reference-distance distributions.

¹²More formally, think of two TADs A and B. Here A consists of k_A and B k_B elements (securities), $P_A = \{p_A(1), \dots, p_A(k_A)\}$ and $P_B = \{p_B(1), \dots, p_B(k_B)\}$, where $k_B > k_A$. Moreover, for simplicity but without loss of generality, assume that $p_A(i) = p_A(j)$ for all $i, j \leq n_A$ and that $p_B(i) = p_B(j)$ for all $i, j \leq n_B$, i.e. the TADs are flat.

Under these circumstances, $H(P_B) > H(P_A)$, but, on the other hand, $H(P_B(k_A)) = H(P_A)$, where $P_B(k_A)$ denotes a truncated TAD at k_A (which is re-normalized). It is clear that under these settings, investor B's strategy is to concentrate less, and thus trade on more securities. However, the use of the truncated TAD suggests the opposite, being then misleading.

Moreover, by assuming that the TADs are based on the trading volumes and that markets are frictionless, one can think that $p_A(k_B) = \epsilon$, that is, $P_A = \{p_A(1), \dots, p_A(k_A), \epsilon\}$, with $\epsilon \downarrow 0$. In other words, investor A *decides* to trade the k_B th security infinitesimal little. In this case, $\lim_{\epsilon \downarrow 0} H(\{p_A(1), \dots, p_A(k_A), \epsilon\}) = H(\{p_A(1), \dots, p_A(k_A)\})$, and therefore the distance to p_B does not change. Therefore, instead of truncating a shorter TAD, one can think that it could be extended by ϵ 's of having as many elements as the longest TAD, and yet the results do not change. This is important, given that we assume that it is the investor's choice not to trade as many securities as the other investors (or herself in the future) trade.

One could ask that, given that the number of securities in the trading list has increased

5. Discussion about trading signatures and investor attention

Investor attention in stock markets has been extensively studied from different points of view, including (Hirshleifer et al. 2009, Pantzalis and Ucar 2014) on limited attention and investor distraction. There are existing papers that measure investors' overall attention based on headlines (Barber and Odean 2007), specific news sources (Barber and Odean 2007, Yuan 2008), search engines (Da et al. 2011), aggregated market data, such as extreme returns (Barber and Odean 2007), upper price limits (Seasholes and Wu 2007), and trading volume (Barber and Odean 2007, Gervais et al. 2001, Hou et al. 2009). However, these proxies measure attention at the aggregate level. A welcome exception is (Gargano and Rossi 2018), who have unique data on investors' browsing history on the brokerage account website, which they used to construct measures of attention at the individual investor level. However, unfortunately, such data are not generally available, and for that reason, these measures are not likely to be commonly used.

As Barber and Odean (2007) puts, "If an unusual number of investors trade a stock, it is nearly tautological that an unusual number is paying attention to that stock". Even if the TAD is not necessarily a good proxy to measure investor attention in stock markets, mainly because a trade event does not always require the investor's attention (this can be the case, e.g., with algorithmic trading) and attention does not always lead to a trade, our results on TADs can still be expected to reflect mechanisms of investor attention.

Investor activity in stock markets are distracted by various things, such as religious holidays (Pantzalis and Ucar 2014), flu waves (McTier et al. 2013), pollen counts (Pantzalis and Ucar 2018), NCAA basketball tournament (March Madness) (Drake et al. 2016), and other value-irrelevant events. On the other hand, these distracting events are, at least to some extent, systemic, i.e., a large portion of investors can simultaneously and systematically react to them. Therefore, such distracting events should not be that crucial to be controlled for when analyzing differences between individual investors. However, in principle, the proxy introduced in this paper could be used to analyze how homogeneously investors react to distracting

over time does the number of securities available in the markets always exceed the length of investor's longest TAD? Indeed, this is the case: the number of securities traded at a given time has always been greater than the size of the portfolio of any investor at any time. From this point of view, there is no need to truncate TADs also with the persistence analysis.

events, which we leave for future research.

6. Conclusion

This paper introduces a novel concept, *trade allocation (frequency-rank) distributions*, TADs, which provide an entirely new way of measuring investor trade allocation patterns in financial markets. Our results establish the following findings:

- (i) Our observations of highly heterogeneous TADs provide evidence that both retail investors and financial institutions allocate their trading efforts heterogeneously among their individual stocks or sub-portfolios, reflecting different strategies. This result can be understood by mental accounting, according to which decision makers can segregate the different types of gambles faced into separate accounts (see Shefrin and Statman 1985, Das et al. 2010). Large variation in an investor's trading intensity across her securities could be explained by bipartite (or multipartite) strategies: for example, by simultaneously acting as a long-term investor with a relatively rarely balanced sub-portfolio and, on the other hand, actively trading other securities in a speculative manner.
- (ii) TADs are distinctive between investors. Surprisingly, TADs are more diverse between financial institutions than between retail investors. This indicated that institutions have distinctive strategies (in terms of on how much different securities are traded) even if the strategies are implemented by multiple traders and despite of regulation and portfolio constraints that are effective especially regarding the insurance companies.
- (iii) TADs are persistent in time with statistical significance. It takes several years (8 to 16) for an investor to exhibit a significantly different TAD. This provides evidence that investors have different mechanisms by which they distribute trading intensity across different securities and apply their mechanisms in a determined and persistent way.
- (iv) Moreover, TAD shapes are not statistically affected by the turnover of securities in the investor's portfolio. Similarly to social relationships where people preserve the distribution of attention across friends even when their social circles changes (Saramäki et al. 2014), the way how investors distribute their trading intensity is not dependent on the set

of securities. This suggests that investors do not ‘ earmark ’ securities. That is, even if investors have security-specific strategies, the securities can be replaced while the strategies remain.

- (v) The larger the set of an investor’s securities, the more frequently each security is traded on average. We do not observe a saturation with large portfolio sizes. This holds equally at the population and the individual level. Our findings suggest that time constraints do not limit investors’ activities; when investors increase their portfolio size, they seem to be able to adjust their strategies accordingly.
- (vi) Investors trade securities in a bursty way, with series of short bursts of trades interspersed with longer gaps.

Instead of studying the rationally-behaving *homo economicus*, we have focused on the *homo sapiens* who conducts trading. Our findings show the importance of the human factor in the analysis financial markets. Our observations on the parallels between financial and social behavior and the computational social science methodology adapted for investor-centric financial systems pave the way for future studies in behavioral finance. Most importantly, TADs have potential applications in studying investor activity and performance in stock markets. There is evidence that active investors earn higher returns than inactive ones (Dahlquist et al. 2017), and it would be interesting to further analyze the relation of the patterns of investors’ TADs to their returns.

Appendix

In principle, the TADs can be calculated using only the information about the buy (sell) transactions. However, we have observed that the differences between average population TADs calculated from different type transactions are minimal, see Figure 5A. The fraction of trading intensity to top alters is summarized in Table 4. When looking at differences between buy and sell TADs at individual level we also observe only marginal differences. For half of the TADs the difference between the fraction of trading intensity dedicated to the top ranking alter in buy and sell TAD ranges between 5.7% to -4.5%, see Figure 5B. For 80% of the TADs the same difference ranges between -16.7% to 16.3 %.

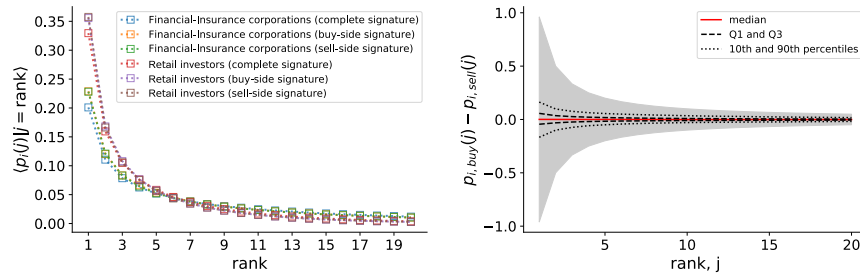


Figure 5: (A) Averaged annual TADs for retail investors and financial institutions for the top 20 ranks. Three TADs are shown for each investor group: TADs created from complete set of transactions, only buy side and only sell side transactions. (B) Change in trading intensity for alters between buy and sell TADs for individual investors.

Investor group	$\left\langle \sum_{n=1}^X P_i^t(n) \right\rangle$	SELL		BUY		ALL		Difference	
		mean	std	mean	std	mean	std	SELL	BUY
Financial/Insurance	$X = 1$	0.228	0.222	0.229	0.228	0.201	0.210	0.027	0.028
	$X = 5$	0.549	0.269	0.552	0.268	0.504	0.265	0.044	0.048
Retail investors	$X = 1$	0.358	0.263	0.356	0.251	0.330	0.246	0.028	0.026
	$X = 5$	0.762	0.226	0.765	0.216	0.728	0.226	0.033	0.037

Table 4: Fraction of trading intensity directed to trading top alters. Third column indicates the fraction of trading intensity for the top and top five alters in sell TADs, fourth column – buy TADs and fifth column – for TADs from all transactions.

In Section “3.1. A small number of securities receive a large share of an investor’s trading intensity”, we analyze the trading patterns from the point of view of the TADs. Here we additionally show in Figure 6 that the fraction of the trading intensity of retail investors allocated to the top traded ranks decreased considerably from 1995 to 2000, but from 2006 onwards

it remained stable. At the same time, institutional investors showed very similar level of trading intensity throughout the period.

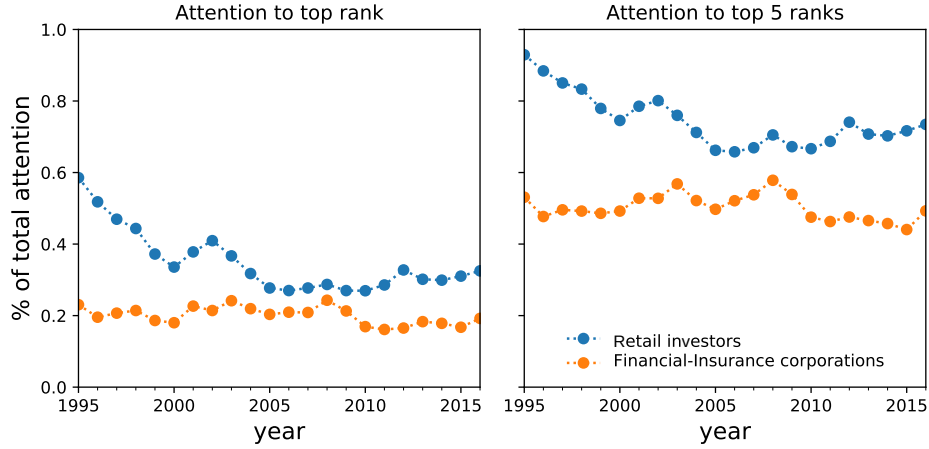


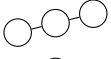
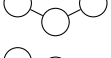
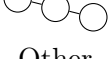
Figure 6: (A) The mean trading intensity allocated to the first ($\langle p_i(1) \rangle$) and (B) to the top five traded securities ($\langle \sum_{j=1}^5 p_i(j) \rangle$).

Regarding Section “3.4 Trading intensity increases with the number of traded securities”, we provide further evidence that investor trading intensity per security increases with the number of securities. In Table 5, for each retail investor i that had at least 3 unique observations for the number of securities traded in a year k_i^t , we form three buckets for the unique observations on k_i . The first bucket is associated to the observations in the years when a given investor has traded the least number of different securities, while the third is the bucket associated to the observations in the years when the investor has traded the most securities. Next, we calculate the median average traded intensity w_i for each bucket. We have four prototypes for the shape of the trading intensity per security with respect to the buckets. From the table we can see that for over 56% of investors have an increasing median average trading intensity for the three buckets. Moreover, less than 30% percent of investors have a saturation point.

6.1. Robustness check with alternative proxies for trading intensity

Figure 7 shows the main results with three alternative measures of trading intensity for both retail investors (Panel A) and financial institutions (Panel B). In both of the panels, the first row shows the results for the measure based on the number of distinct trading days used in the main paper,

Table 5: Distribution of different $w_i \sim f(k_i)$ curve shapes among retail investors and financial-insurance corporations.

shape	Retail investors		Fin.-Ins. corp.	
	N	%	N	%
	157 898	56.12	343	43.75
	73 813	26.24	235	29.97
	34 123	12.13	153	19.52
	9 330	3.32	48	6.12
Other	6 185	2.20	5	0.64
Total	281 349	100.00	784	100.00

the second row for the alternative measure based on the number of transaction, and the third row on the total traded volume. Regarding the second last measures, after 2009 our data sets does not report each transactions, but the transactions are pooled over the trading days. For that reason, in results reported in Fig 7 is based on data between 1995–2009 (and not 1995–2016 like in the main paper).

The first column reports averaged annual TADs in the same way as reported in Fig. 1B in the main paper. We see that the average TADs are very similar across different trading intensity measures, indeed. The second column shows the distributions of self- and reference-distances, d_{self} and d_{ref} , respectively, in relation to Fig. 2A in the main paper. Again, the results are very robust from the point of view of alternative measures. Some variation between the measures can be observed with the plots in the third column that show the mean self-distances and reference-distances for different time lags (see Fig 2B in the main paper). However, the conclusion remains the same: investors TAD are very persistent and it takes several years until the TADs become statistically different. The fourth column is in relation to Fig. 3A in the main paper about investors' average trading intensity with respect to the number of securities traded (size of the portfolio). In Panel B, we observe a downward trend at the beginning, resulting an asymmetric U-shape, which we do not observe in Fig. 3A in the main paper. This is because the data span over different horizon than in the main paper (1995–2016 vs 1995–2009) for the calculation of alternative measures. Nevertheless, each security on average is traded the most in the case of the largest portfolios. Overall, we observe that the results are very consistent across alternative

measures.

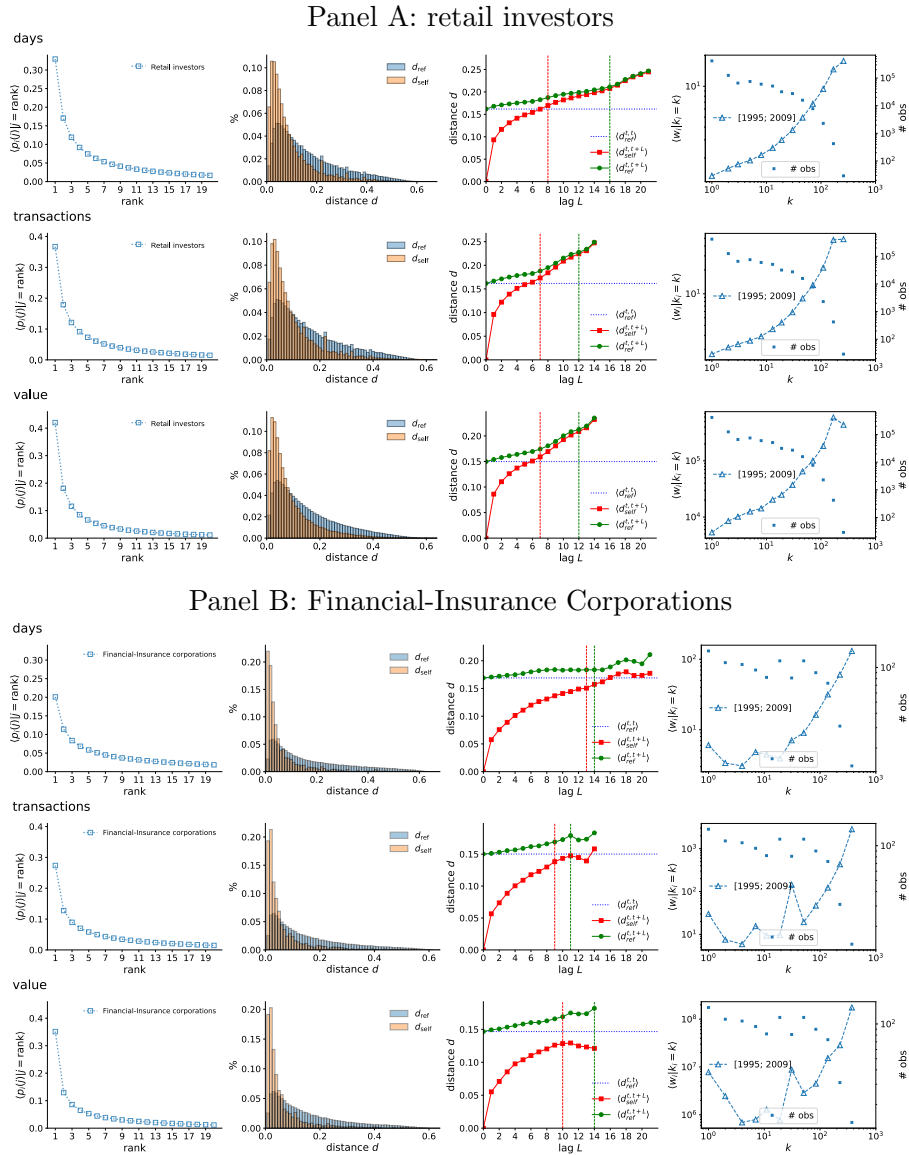


Figure 7: Alternative measures for TADs. Panel A shows the results for retail investors and Panel B for financial institutions. In both of the panels, the first row shows the results for the measure based on the number of distinct trading days used in the main paper, the second row for the alternative measure based on the number of transaction, and the third row on the total traded volume.

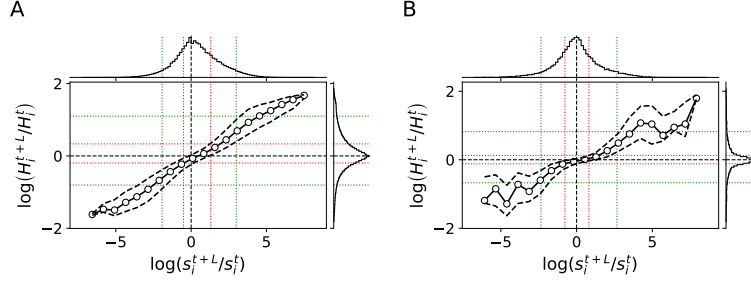


Figure 8: (A) – retail investors, (B) – Financial-Insurance corporations. Relationship between the change in total trading intensity and the median change in TAD entropy. Black dashed lines around the median: first and third quartiles. Vertical and horizontal lines.

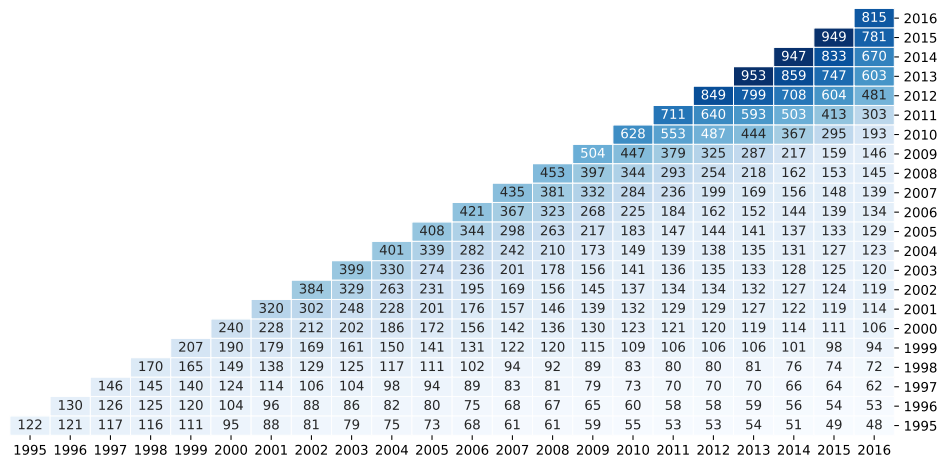
6.2. Additional analysis: Increase in trading intensity means flatter TADs

Figure 3 in the main paper suggests a positive relationship between the average trading intensity and the number of traded securities. For that reason, we ask how investors redistribute their trading intensity among their securities when they increase or decrease their overall trading intensity s_i . We measure the relative change in overall trading intensity between periods t and $t+L$ with $\log(s_i^{t+L}/s_i^t)$. For the measure of re-distribution, we use the log of TAD entropy ratio— $\log(H(P_i^{t+L})/H(P_i^t))$, where H is the entropy measure,

$$H(P_i^t) = - \sum_{r=1}^{k_i} p_i(r) \log p_i(r), \quad (7)$$

where $p_i(r)$ is the fraction of total trading intensity directed to security r , k_i is the number of securities traded by i , and, P_i^t is investor i 's TAD in period t . The TAD entropy is equal to 0 when all trading intensity is concentrated on one security, and for a given number of traded securities k_i it is highest when the trading intensity is distributed equally to all securities.

Figure 8 shows that an increase/decrease in total trading intensity is associated to an increase/decrease in TAD entropy. This means that if an investor becomes more active, her distribution of trading intensity on the securities, *i.e.*, the TAD, becomes more uniform. This can happen in two ways: i) either the investor starts trading more securities and/or ii) she distributes her intensity more evenly to the traded securities.



6.3. Additional statistics

6.4. Tabulated data

Panel A. Illustration of TAD construction.															
Figure 1A rank/j	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
w_{ij} (non-sorted)	12	22	5	15	8	13	9	11	10	25	8	14	7	19	3
w_{ij} (sorted)	25	22	19	15	14	13	12	11	10	9	8	8	7	5	3
$p_i(rank)$	0.14	0.12	0.10	0.08	0.08	0.07	0.07	0.06	0.06	0.05	0.04	0.04	0.04	0.03	0.02
Panel B. Mean TADs															
Figure 1B rank	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Financial-Insurance	0.20	0.11	0.08	0.07	0.06	0.05	0.04	0.04	0.04	0.03	0.03	0.03	0.03	0.03	0.02
Households	0.33	0.17	0.12	0.09	0.07	0.06	0.05	0.05	0.04	0.04	0.03	0.03	0.03	0.03	0.02
Panel C. Signature slope CDFs															
Figure 1C x, slope	0.02	0.09	0.16	0.23	0.30	0.37	0.44	0.51	0.58	0.65	0.72	0.79	0.86	0.93	1.00
Financial-Insurance	0.77	0.89	0.92	0.94	0.95	0.96	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	1.00
Households	0.47	0.78	0.84	0.87	0.90	0.92	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	1.00

Table 6: Tabulated data for Figure 1. Row " w_{ij} (non-sorted)" indicates the values used to illustrate the investor trade allocation graph in the inset of Fig. 1A. The numbers in rows " w_{ij} (sorted)" and " $p_i(rank)$ " correspondingly show the sorted and normalized trading intensities used to illustrate the construction of a TAD in Fig. 1A. The numbers in Panel B indicate the TAD values for the first 15 ranks. Panel C tabulates the CDF of the TAD slope distributions for selected slope values. In Fig. 1C, the vertical lines indicate the mean slopes for Financial-Insurance corporations at 0.06 and retail investors at 0.12.

Panel A. Self and reference distance distribution.																						
Figure 2A distance d	0.01	0.04	0.07	0.1	0.13	0.16	0.19	0.22	0.25	0.28	0.31	0.34	0.37	0.4	0.43	0.46	0.49	0.52	0.55	0.58	0.61	0.64
d_{self}	0.066	0.094	0.057	0.037	0.022	0.015	0.011	0.018	0.0058	0.0063	0.0028	0.0024	0.0022	0.0015	0.0019	0.0011	0.00061	0.00029	0.00012	2e-05	2e-05	0
d_{ref}	0.014	0.051	0.046	0.039	0.031	0.027	0.022	0.023	0.016	0.014	0.011	0.0092	0.008	0.0066	0.006	0.0044	0.0033	0.0023	0.0012	0.00035	4.9e-05	7.8e-07
Figure 2D distance d	0.01	0.04	0.07	0.1	0.13	0.16	0.19	0.22	0.25	0.28	0.31	0.34	0.37	0.4	0.43	0.46	0.49	0.52	0.55	0.58	0.61	0.64
d_{self}	0.22	0.086	0.041	0.022	0.012	0.0047	0.007	0.009	0.002	0.003	0.001	0.00067	0	0.001	0.0037	0.001	0.00033	0.00033	0.001	0.00067	0	0
d_{ref}	0.023	0.055	0.042	0.033	0.025	0.021	0.018	0.017	0.014	0.013	0.011	0.0099	0.0087	0.0079	0.0072	0.0065	0.0053	0.0042	0.0035	0.0028	0.0011	0.00032
Panel B. Decay in TAD similarity.																						
Figure 2B lag L	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
$\langle d_{\text{ref}}^{t,t} \rangle$	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16
$\langle d_{\text{self}}^{t,t+L} \rangle$	0	0.093	0.12	0.13	0.14	0.15	0.15	0.16	0.17	0.18	0.18	0.19	0.19	0.19	0.2	0.2	0.21	0.22	0.23	0.23	0.24	0.24
$\langle d_{\text{ref}}^{t,t+L} \rangle$	0.16	0.17	0.17	0.17	0.18	0.18	0.18	0.18	0.19	0.19	0.19	0.2	0.2	0.2	0.2	0.21	0.21	0.22	0.23	0.24	0.24	0.25
Figure 2E lag L	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
$\langle d_{\text{ref}}^{t,t} \rangle$	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17
$\langle d_{\text{self}}^{t,t+L} \rangle$	0	0.058	0.076	0.089	0.1	0.11	0.12	0.13	0.13	0.14	0.14	0.14	0.15	0.15	0.16	0.16	0.17	0.18	0.18	0.17	0.17	0.18
$\langle d_{\text{ref}}^{t,t+L} \rangle$	0.17	0.17	0.17	0.17	0.17	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.19	0.2	0.2	0.2	0.19	0.21
Panel C. Decay in Jaccard similarity.																						
Figure 2C lag L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	
all	0.35	0.24	0.19	0.16	0.14	0.13	0.12	0.1	0.092	0.083	0.076	0.069	0.064	0.058	0.052	0.048	0.042	0.035	0.03	0.027	0.024	
top 10	0.29	0.2	0.17	0.14	0.12	0.11	0.1	0.092	0.083	0.076	0.071	0.066	0.062	0.056	0.051	0.048	0.042	0.035	0.031	0.029	0.026	
top 5	0.25	0.17	0.14	0.12	0.1	0.096	0.088	0.08	0.074	0.069	0.065	0.062	0.058	0.054	0.049	0.045	0.041	0.034	0.03	0.029	0.029	
top 3	0.23	0.16	0.13	0.11	0.094	0.088	0.081	0.075	0.07	0.067	0.064	0.062	0.058	0.053	0.048	0.044	0.04	0.035	0.03	0.03	0.031	
Figure 2F lag L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	
all	0.46	0.35	0.29	0.25	0.22	0.21	0.19	0.17	0.16	0.14	0.13	0.12	0.11	0.099	0.091	0.084	0.079	0.07	0.064	0.061	0.049	
top 10	0.3	0.22	0.19	0.16	0.14	0.13	0.12	0.11	0.1	0.095	0.09	0.083	0.077	0.069	0.063	0.056	0.05	0.047	0.044	0.046	0.03	
top 5	0.25	0.18	0.15	0.13	0.12	0.11	0.1	0.09	0.081	0.076	0.073	0.066	0.063	0.056	0.052	0.046	0.039	0.037	0.032	0.039	0.027	
top 3	0.24	0.17	0.14	0.12	0.11	0.11	0.099	0.086	0.078	0.073	0.069	0.065	0.062	0.053	0.053	0.041	0.039	0.039	0.033	0.03	0.031	

Table 7: Tabulated data for Fig. 2.

Panel A. Average trading intensity with respect to the number of securities traded.																					
Figure 3A		1	2	3	5	8	13	19	30	46	71	110	169	259	398						
k																					
$\langle w_i k_i = k \rangle$ in 1995 – 1999	1.3	1.4	1.5	1.7	1.9	2.1	2.5	3	3.8	4.8	7.1	9.5	-	-	-						
$\langle w_i k_i = k \rangle$ in 1995 – 2016	1.4	1.7	1.8	2	2.3	2.7	3.3	4.2	5.6	7.8	11	16	21	33	-						
$\langle w_i k_i = k \rangle$ in 2012 – 2016	1.7	1.7	1.8	2	2.3	2.8	3.4	4.5	6.3	9.3	13	16	27	-	-						
Figure 3C		1	2	4	7	11	19	31	51	84	138	227	372	610	1000						
k																					
$\langle w_i k_i = k \rangle$ in 1995 – 1999	2.1	1.7	2.2	2.2	2.1	2.9	4.7	7.5	12	34	63	-	-	-	-						
$\langle w_i k_i = k \rangle$ in 1995 – 2016	7.7	3.8	6.3	7.1	5.2	5.5	8	11	15	28	46	1.1e+02	1.8e+02	1.8e+02	-						
$\langle w_i k_i = k \rangle$ in 2012 – 2016	7.9	8.8	6.8	5.3	4.5	5.9	5.4	13	23	17	39	46	56	2.3e+02	-						
Panel B. Change in the number of traded securities vs change in the average trading intensity.																					
Figure 3B		-3.5	-3.1	-2.7	-2.3	-1.9	-1.5	-1.1	-0.67	-0.27	0.14	0.54	0.94	1.3	1.8	2.2	2.6	3	3.4	3.8	4.2
$\log(k_i^{t+L}/k_i^t)$																					
median $\log\left(w_i^{t+L}/w_i^t\right)$		-1.6	-1.1	-0.95	-0.73	-0.57	-0.41	-0.26	-0.13	-0.018	0.085	0.22	0.4	0.55	0.69	0.83	0.96	1.1	1.4	1.7	1.6
Q1 $\log\left(w_i^{t+L}/w_i^t\right)$		-1.8	-1.6	-1.4	-1.2	-1	-0.88	-0.72	-0.54	-0.38	-0.25	-0.13	0	0.12	0.25	0.38	0.51	0.67	0.87	1.1	1
Q3 $\log\left(w_i^{t+L}/w_i^t\right)$		-1.2	-0.61	-0.43	-0.19	0.0055	0.16	0.26	0.33	0.38	0.46	0.64	0.86	1	1.2	1.3	1.5	1.6	1.9	2.2	2.5
Figure 3D		-3.3	-2.9	-2.5	-2.1	-1.7	-1.2	-0.83	-0.41	0.00058	0.42	0.83	1.2	1.7	2.1	2.5	2.9	3.3	3.7	4.2	4.6
$\log(k_i^{t+L}/k_i^t)$																					
median $\log\left(w_i^{t+L}/w_i^t\right)$		-0.85	-0.96	-0.82	-0.76	-0.75	-0.57	-0.45	-0.29	-0.059	0.094	0.27	0.43	0.49	0.59	0.51	0.59	0.52	0.67	1.8	2.9
Q1 $\log\left(w_i^{t+L}/w_i^t\right)$		-1.3	-1.6	-1.2	-1.1	-1.4	-1.1	-0.93	-0.75	-0.45	-0.27	-0.16	-0.0042	0.036	0.18	-0.0076	0.19	0.16	0.044	0.93	2.6
Q3 $\log\left(w_i^{t+L}/w_i^t\right)$		0.021	0.12	-0.17	-0.23	-0.27	-0.081	0.059	0.15	0.33	0.5	0.71	0.91	0.98	1.1	1.1	1	0.86	1.1	2.9	3.3

Table 8: Tabulated data for Fig. 3.

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